National University of Computer and Emerging Sciences

CS3009 — Software Engineering | Spring 2026

Team Information

Deliverable 3 — Final Sprint

Ibwaan

(Project: Naawbi — Learning Management System)

Team Lead	**Ibraheem Farooq**
Members	23i-0708 | Ali Aan Khowaja 23i-0816 | Ibraheem Farooq 23i-0819 | Wajih-Ur-Raza Asif
Submission Date	20 April 2026
Repository	github.com/aliaankhowaja/Naawbi

1 Team Updates for Sprint 3

Status summary

No changes to team membership or roles.
Team Lead confirmed: Ibraheem Farooq continues as Team Lead.
Team agreement: the agreement established in Deliverable 1 remains fully active for Sprint 3. No revisions to communication methods, meeting schedules, version control practices, work division, submission responsibilities, or contingency planning were required.

- **Team membership:** unchanged since Deliverable 1. All three original members carried through to the final sprint.
- **Development: all three members contribute to development equally** across every sprint — this has been our working model from Sprint 1 onwards and continues unchanged in Sprint 3.
- **Roles & responsibilities:** the additional (non-development) ownership areas are unchanged in substance. On top of the shared development work, Ibraheem leads authentication/RBAC + attachments, Wajih leads assignments tab + grading/gradebook + test plan, and Ali leads Sprint 2 features + Sprint 3 UI screenshots. Detailed in the Deliverable 3 document, Section "Work Division for Sprint 3".
- **Communication:** WhatsApp + in-person + Google Meet — unchanged.
- **Meeting cadence:** 4 check-ins per week — unchanged. Sprint 3 planning held 26 March; mid-sprint review 8 April; sprint close retrospective 20 April.
- **Version control:** same branch-per-feature + PR-review policy, atomic commits with `feat:`/`fix:`/`docs:`/`chore:` prefixes — unchanged.
- **Submission responsibilities:** Wajih remains submission owner; Ibraheem and Ali review before submission; internal cutoff is 24 hours before the deadline — unchanged.
- **Contingency planning:** unchanged. Scope-down-then-redistribute protocol did not need to be invoked during Sprint 3.

2 Project Team Information

Team Name	Ibwaan
Project Name	Naawbi
Team Lead	Ibraheem Farooq
Deliverable	Deliverable 3 (Sprint 3 / Final Sprint)

2.1 Team Roles (Unchanged from D1)

Member	Role(s)	GitHub
Ibraheem Farooq	PM, Scrum Master, Developer	github.com/ibbi1020
Wajih-Ur-Raza Asif	Requirements Architect, Tester, Developer	github.com/wajih-rathore
Ali Aan Khowaja	Developer, UI Designer	github.com/aliaankhowaja

2.2 Member Biographies

Ibraheem Farooq 23i-0816

PM / Scrum Master / Developer

A Computer Science major based in Islamabad who transitioned from designing posters and brand identities to UX and product design. His technical background provides a unique perspective on bridging digital experiences that balance user needs with technical viability. Passionate about solving real problems through thoughtful design and clean, intuitive interfaces.

Wajih-Ur-Raza Asif 23i-0819

Requirements Architect / Tester / Developer

A software engineer focused on building practical systems that solve real operational problems. His background spans end-to-end projects across different technologies, with a strong interest in automation, AI-driven workflows, and scalable system design. Currently expanding his understanding of enterprise technology and systems that make processes more efficient and reliable.

Ali Aan Khowaja 23i-0708

Developer / UI Designer

An AI Engineer driven by a simple goal: building applications that work at scale. His expertise lies at the intersection of high-performance backend architecture and agentic systems. He specialises in designing and deploying scalable infrastructures that power intelligent applications. Beyond the code, he is a robotics enthusiast passionate about bringing new ideas to life.

2.3 Team Agreement (Active Through Sprint 3)

The agreement below was established in Deliverable 1 and remains active for Sprint 3 without modification. It is reproduced here for completeness.

Communication

- **WhatsApp** — primary channel for daily coordination; respond within 1 hour.
- **In-person** — preferred when on campus.
- **Google Meet** — for remote meetings.
- Urgent issues: tag the team and respond within 20 minutes.
- If unavailable: post an async update in WhatsApp within 24 hours.

Meetings

- Mandatory: Sprint Planning, Sprint Review, Sprint Retrospective.
- Regular check-ins: 4× per week (in-person or Google Meet).
- Absence: notify team 24 hours in advance and share an async update.
- Every meeting ends with: decisions + action items + owners.
- Notes: rotating assigned person.

Meeting Preparation

- **Sprint Planning:** review backlog, propose task breakdowns.
- **Review:** prepare demo flow + screenshots.
- **Retrospective:** each member prepares (1) what went well, (2) what did not, (3) what to change.

Version Control

- Small, atomic commits; new branch per ticket.
- PR required before merging; do not commit secrets, credentials, or large binaries.
- Commit message format: `feat:`, `fix:`, `docs:`, `chore:`

Division of Work (Unchanged Ownership Model)

All three team members contribute to development equally on every sprint. The breakdown below lists additional *non-development* ownership areas on top of each member's equal development contribution.

Owner	Additional Responsibility (on top of equal development)
Wajih	Requirement ownership: user stories, structured specifications, consistency checks, test planning, and test execution.
Ali	UI ownership: screen layout, visual design system, and iteration screenshot production.
Ibraheem	Coordination: sprint tracking, board management, packaging, and final submission.

Contingency Plan. If a member becomes unavailable or falls behind, the team re-scopes to the minimum required deliverable content, redistributes work immediately, and prioritises required artefacts. The PM/SM updates the board, assigns new owners, and schedules an urgent meeting to unblock progress. *Not invoked during Sprint 3.*

Submission Protocol (Unchanged)

- **Submission owner:** Wajih.
- **Internal review:** Ibraheem and Ali.
- **Internal cutoff:** finalise 24 hours before the deadline.